RSSpad • Podcast & RSS 2.0 Feed Studio • User Manual & Reference

Linux Mint Cinnamon Desktop Integration • Automated GitHub Releases + GitHub Pages Hosting

◆ Overview & Core Architecture

RSSpanel is a lightweight GTK 3 desktop studio that converts any folder of video or audio tracks into an Apple Podcasts / iTunes-compliant RSS 2.0 feed and modern streaming web portal. Large media files bypass GitHub's 100 MB Git blob restriction by uploading directly to a GitHub Release, while the lightweight XML feed and responsive HTML player are pushed to GitHub Pages.

[CORE HIGHLIGHTS] Multi-format scanning (.mp4, .m4v, .mov, .mp3, .wav, .flac, .ogg, .m4a) • Automatic video poster frame generation • Native GStreamer media player • Local Wi-Fi testing server • Background file watcher • PythonPanel dark neon design.

◆ Daily Publishing Workflow

• **1. Drop Media:** Save or export your finished .mp4, .wav, or .mp3 track directly into your feed directory (or drag-and-drop into RSSpanel).

• **2. Auto-Detect:** RSSpanel calculates runtime, size, auto-generates a clean video preview frame, and indexes the episode instantly.

• **3. One-Click Publish (F6):** Press F6 or click 'Generate & Publish'. New episodes upload to GitHub Releases, and feed.xml + index.html update on Pages.

• **No Terminal Needed:** Runs completely hands-off. Podcast subscribers receive new tracks on their next refresh.

◆ Interface Tour & Toolbar Controls

• **Folder (Ctrl+O):** Switch between feed directories or open an existing audio/video show.

• **Add Tracks (Ctrl+I):** Import single or batch media files directly into the active feed folder.

• **Rescan (Ctrl+R):** Refresh folder contents and recalculate durations/sizes.

• **Generate Feed (F5):** Compile feed.xml and index.html locally without uploading.

• **Publish (F6):** Upload new episodes to GitHub Releases and push updates to GitHub Pages.

• **LAN Server (Toggle):** Starts Wi-Fi HTTP preview server (http://<lan-ip>:8765) for local phone testing.

• **Player (Space):** Built-in GStreamer engine with status bar progress and play/pause.

• **Episode Editor:** Select any track in the list to customize show title & description.

◆ Studio Workspace Panels

• **Tracks Tab (Ctrl+1):** View complete track table with status icons, episode titles, filenames, formats, durations, sizes, and timestamps.

• **feed.xml Tab (Ctrl+2):** Live GtkSourceView syntax-highlighted preview of the exact XML document that podcast clients read.

• **Activity Log (Ctrl+3):** Real-time terminal log showing FFmpeg conversions, poster extraction, release uploads, and git push outputs.

• **Media Tools Menu:** Menu → → Media & Cover Tools allows extracting square cover art (cover.jpg) or batch converting tracks to 192k MP3.

◆ Keyboard Shortcuts Quick Reference

Shortcut	Action Description
F5 / Ctrl + S	Generate feed.xml & index.html locally
F6	Generate & Publish to GitHub Releases + Pages
Ctrl + O	Open a podcast / media channel folder
Ctrl + I	Import tracks into the active feed folder
Ctrl + R	Rescan folder and update media durations
Ctrl + B / F9	Toggle Feed Settings sidebar
Space	Play or pause selected audio/video track
Delete	Rename track or move file to Trash
Ctrl + Shift + C	Copy public feed.xml URL to clipboard
Ctrl + 1 / 2 / 3	Switch between Tracks, feed.xml, and Activity Log tabs
Ctrl + Q	Quit RSSpanel

◆ Configuration Schema (.rsspanel.json)

Channel configuration is saved in an unobtrusive **.rsspanel.json** file inside your media directory:

- **base_url:** Public base URL for feed and web player (e.g. https://<user>.github.io/<repo>).
- **github_repo:** Target repository ('owner/repo') used for GitHub Releases and Pages hosting.
- **github_release_tag:** Release tag name (default: 'episodes') where files over 95 MB are hosted.
- **image_url:** Square podcast show artwork URL (1400x1400 to 3000x3000 px).
- **title / description:** Show title and channel description visible in podcast directories.
- **publish_command:** Optional custom command (e.g. rsync or git) if not using GitHub.

◆ Headless CLI & Automation

[TERMINAL & CRON USAGE] **# Compile feed locally:** python3 feed_core.py /path/to/media/folder
Upload & publish to GitHub: python3 feed_core.py --sync-github /path/to/media/folder
Launch GUI: rsspanel or rsspanel /path/to/media/folder

◆ Linux Mint Cinnamon Integration

- **Panel Shortcut:** Right-click 'RSSpanel' in the Cinnamon Menu (Sound & Video / Favorites) and select 'Add to panel'.
- **Application Location:** Launcher symlinked to ~/.local/bin/rsspanel with full GTK source & icon integration.
- **Desktop Launcher:** A verified desktop entry is maintained at ~/.local/share/applications/rsspanel.desktop for Cinnamon Menu launch.
- **HiDPI Retina Icon:** Vector scalable SVG icon installed across 16px to 512px system icon themes.